USA Housing Rental Price Analysis

Rental price drivers and where a pricing model misses

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0.76	$176	0.00
R-squared (random forest)	Typical error per month	R-squared before cleaning

1. Background and question

What drives rental prices, and where does a pricing model systematically under- or over-estimate rent?

Rental pricing is a trade-off: price too high and units sit vacant, too low and revenue is lost. Owners also decide which amenities to invest in. Listing data from across the U.S. shows what renters actually pay for, and a pricing model can flag units that look over- or under-priced for their market.

2. Data

- USA Housing Listings (Kaggle): 384,977 Craigslist rental listings across 51 states and 404 regions, 22 columns, from the original team capstone (BUAN-685, Sacred Heart University, Nov 2024)
- Variables: price, square feet, beds, baths, property type, pet/smoking/wheelchair/EV/furnished flags, laundry, parking, region, state, coordinates
- Raw data quality problems: half the listings were duplicate reposts, and prices ranged from $0 to $2.8 billion

3. Method

- Reproduced the original course model (linear regression of price on size, beds and baths, raw data) as the starting point
- Cleaning: removed 191,384 duplicates, 2,710 impossible values and 127 listings in tiny regions, leaving 190,756 listings
- Exploratory analysis: price distribution, price vs. size by type, boxplots, correlations, amenity premiums
- Models: mean-rent baseline, linear regression, random forest, gradient boosting; 75/25 split plus cross-validation; R-squared and MAE
- Error analysis: mean residual by property type and state to find systematic under- and over-pricing

4. Project walkthrough

The problem. What drives rental prices across the U.S., and where does a pricing model consistently get prices wrong?

Starting point. I first reran our original course model (linear regression of price on size, beds and baths). It scored R-squared 0.00 with an RMSE of about $10 million, which is a sign of a data problem, not a modelling problem.

Finding the data problems. Half the 384,977 listings were duplicate reposts of the same unit, and prices ranged from $0 to $2.8 billion. A few absurd values were dominating every fit.

Cleaning. I removed exact duplicates, kept rents between $300 and $10,000 and sizes between 200 and 5,000 sq ft, and dropped regions with under 20 listings, leaving 190,756 realistic listings.

Modelling. I compared a mean-rent baseline (typical error $467), linear regression, random forest and gradient boosting, using size, rooms, amenities, laundry, parking, property type, state and coordinates. Random forest was best: R-squared 0.76 and typical error $176 a month.

Error analysis. I then looked at where the model misses. It over-predicts rent in low-cost states (Missouri by about $82) and under-predicts in high-cost states (Massachusetts by about $36). It pulls prices toward the national average.

Why it matters. For owners, the model plus error analysis says which amenities carry a premium (in-unit washer/dryer: about $420 a month at the median; pet-friendly policies: none) and where national pricing rules will mislead them.

5. Results

Model	Test R-sq.	MAE	CV R-sq.
Original course model (raw data)	0.00	RMSE ~$10M	n/a
Baseline (mean rent)	0.00	$467	0.00
Linear Regression	0.48	$303	0.49
Random Forest	0.76	$176	0.71
Gradient Boosting	0.74	$207	0.73



Figure: Rent distribution, rent vs. square footage by type (20K-listing sample), and rent by property type

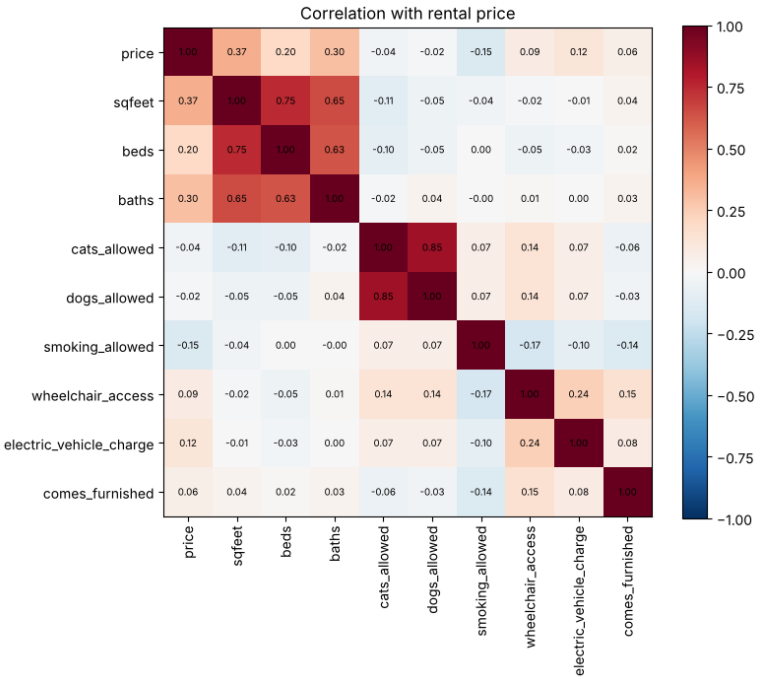


Figure: Correlation of size and amenity flags with rent

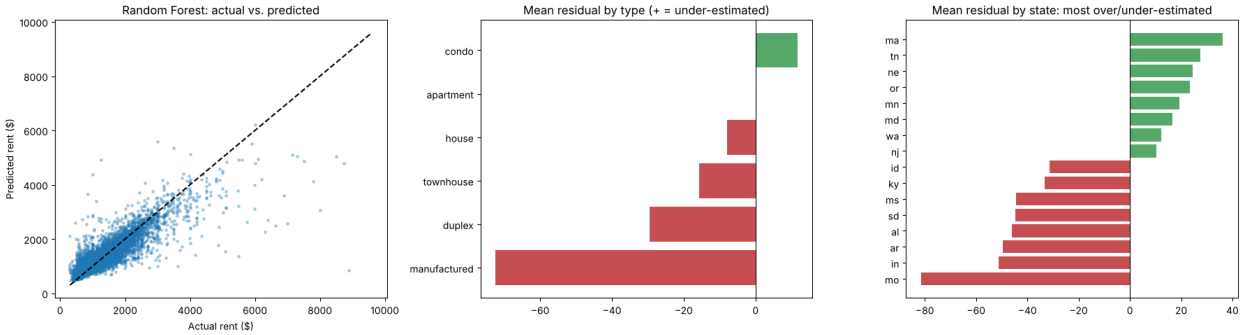


Figure: Actual vs. predicted rent, and mean residual by property type and by state

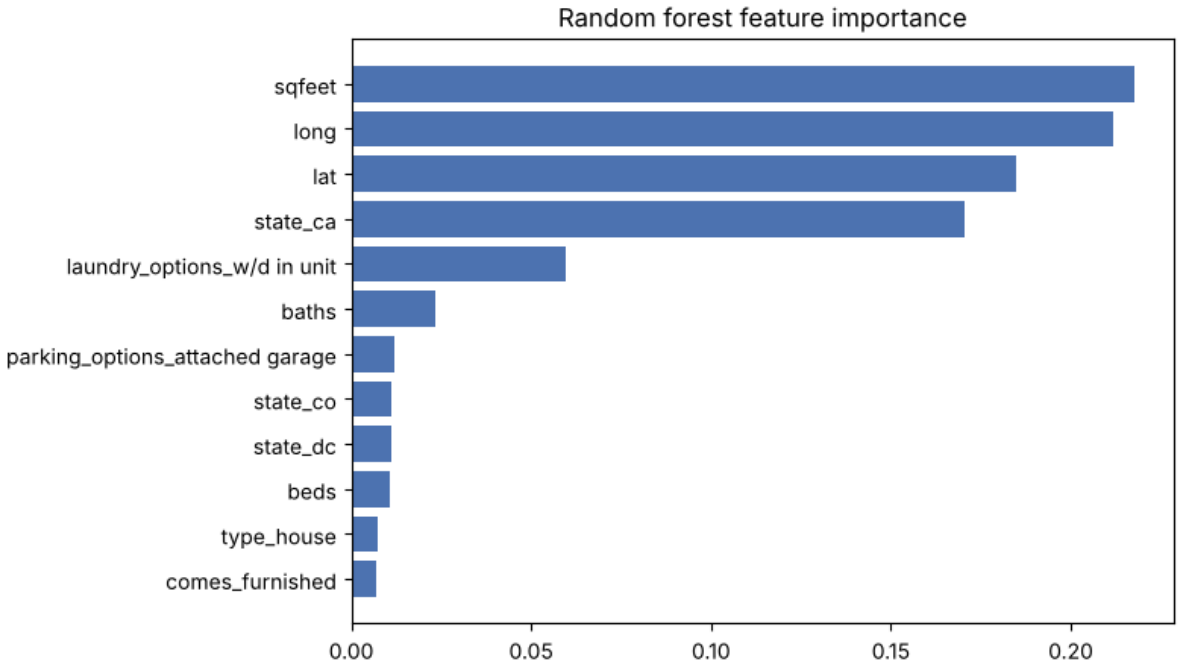


Figure: Random forest feature importance

6. Key findings

- Cleaning turned an unusable model into a useful one: R-squared rose from 0.00 on the raw data to 0.76 with random forest, and typical error fell from $467 to $176 a month.
- Location matters as much as size: square footage, longitude, latitude and being in California are the top four features.
- In-unit washer/dryer is the most valuable common amenity: median rent about $1,370 vs. about $950 with shared laundry. Pet-friendly policies show no premium, matching the team's t-test ($p = 0.32$).
- The model over-estimates low-cost states (Missouri by about $82; Indiana, Arkansas, Alabama by $45-50) and under-estimates high-cost states (Massachusetts, Oregon, Maryland, Washington, New Jersey).
- Manufactured homes are over-estimated by about $72 and duplexes by about $30; apartments are nearly unbiased. The model pulls prices toward the national average.

7. Why these choices

Cleaning before modelling. The jump from R-squared 0.00 to 0.76 came mostly from cleaning. Checking the data before trying fancier models is the main lesson.

Rule-based outlier limits. $300-$10,000 rent and 200-5,000 sq ft are realistic bounds for residential rentals and easy to explain, unlike purely statistical cut-offs.

MAE as the main metric. Mean absolute error is in dollars per month, so it is directly meaningful to a landlord: 'the model is typically off by $176'.

Coordinates as features. Latitude and longitude let tree models learn local price patterns that state alone misses; they became two of the top three features.

8. Key terms explained

R-squared	Share of the variation in rent the model explains: 0 = no better than guessing the average, 1 = perfect.
MAE	Mean absolute error: the average size of the prediction error in dollars.
RMSE	Root mean squared error: like MAE but punishes large errors more, which is why one $2.8 billion listing made it explode.
Residual	Actual rent minus predicted rent. Positive = the model under-estimated; negative = over-estimated.
Random forest	Many decision trees trained on random subsets of the data, averaged together for a stable prediction.
Duplicate reposts	The same unit listed several times; keeping them over-weights some units and leaks near-identical rows between training and test sets.

Tools and skills

Python, pandas, scikit-learn, large-data cleaning, feature engineering, linear regression, random forest, gradient boosting, residual analysis, matplotlib

Code: usa_housing_rental_analysis.ipynb in this folder reproduces every number in this report.